data, and sends it to the originating smart contract. Additionally, the service may offer methods to verify the authenticity of the data fetched.

12.12 From foundations to practical systems

Blockchain is based on nearly 40 years of scientific research in cryptography and math algorithms. The butterfly effect is defined as a little perturbation of an initial state that culminates in significant outcomes. This concept is defined in the context of chaos theory. The advent of Bitcoin has had a butterfly effect on technology. It has creatively synthesized scientific research results from the past four decades to release an innovative working model for a peer-peer digital payment system, as shown in figure 12.10. It shows how foundational research in public-key cryptography and security has led to the discovery of the internet and secure distributed systems. These discoveries, along with hashing and cryptographic algorithms, enabled Bitcoin and the underlying foundation of blockchain. All these concepts culminated in blockchain’s distributed ledger, disintermediation, smart contracts, decentralized apps, and planetary-level inclusive systems.

The smart contract concept in figure 12.10 is enabled by blockchain. A smart contract represents a token. In other words, a smart contract can provide an elegant representation of an asset in the physical world or digital world: a stock, a piece of real estate (as discussed in chapter 9), or a digital game bounty. This tokenized asset can be traded, discounted, divided, archived, and destroyed. The tokens (or their fractions) can be used for incentivization, such as shopping coupons and